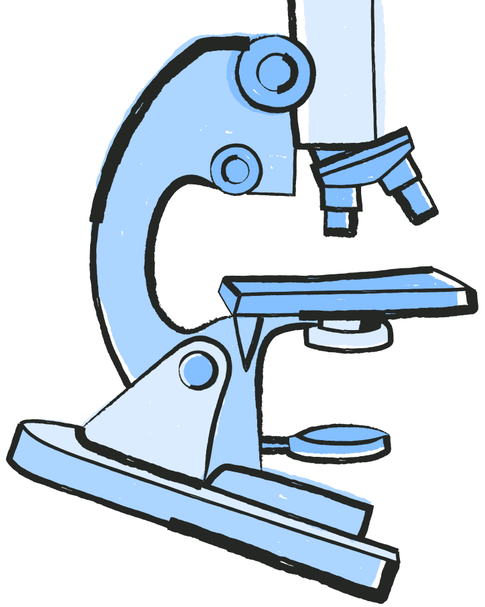
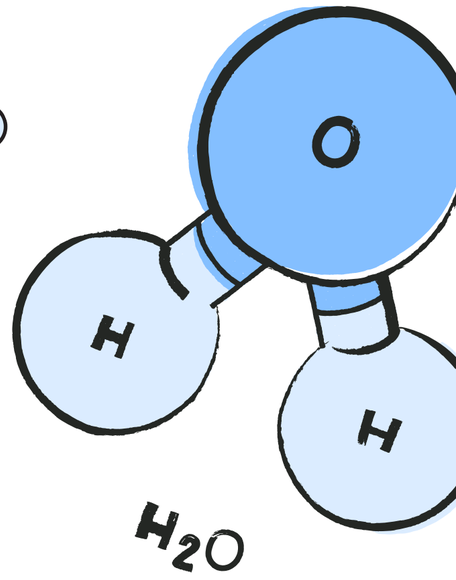
Arowwai Industries

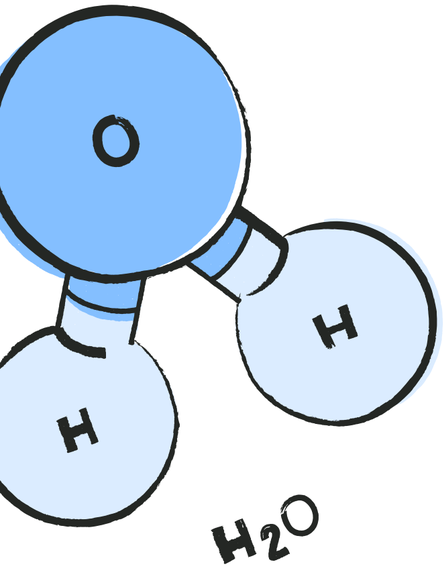
RESEARCH PROPOSAL

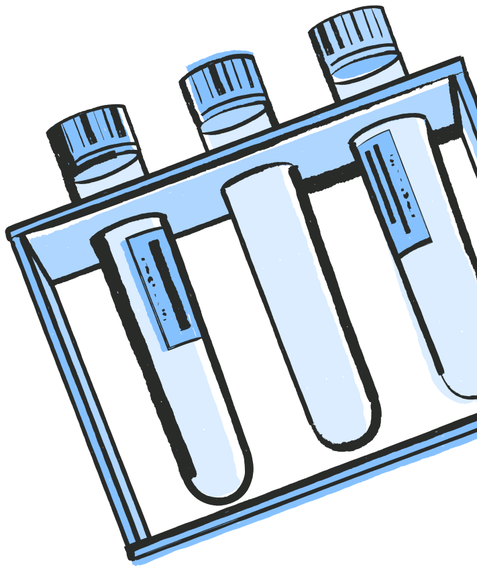
PRESENTATION

March 14, 2027





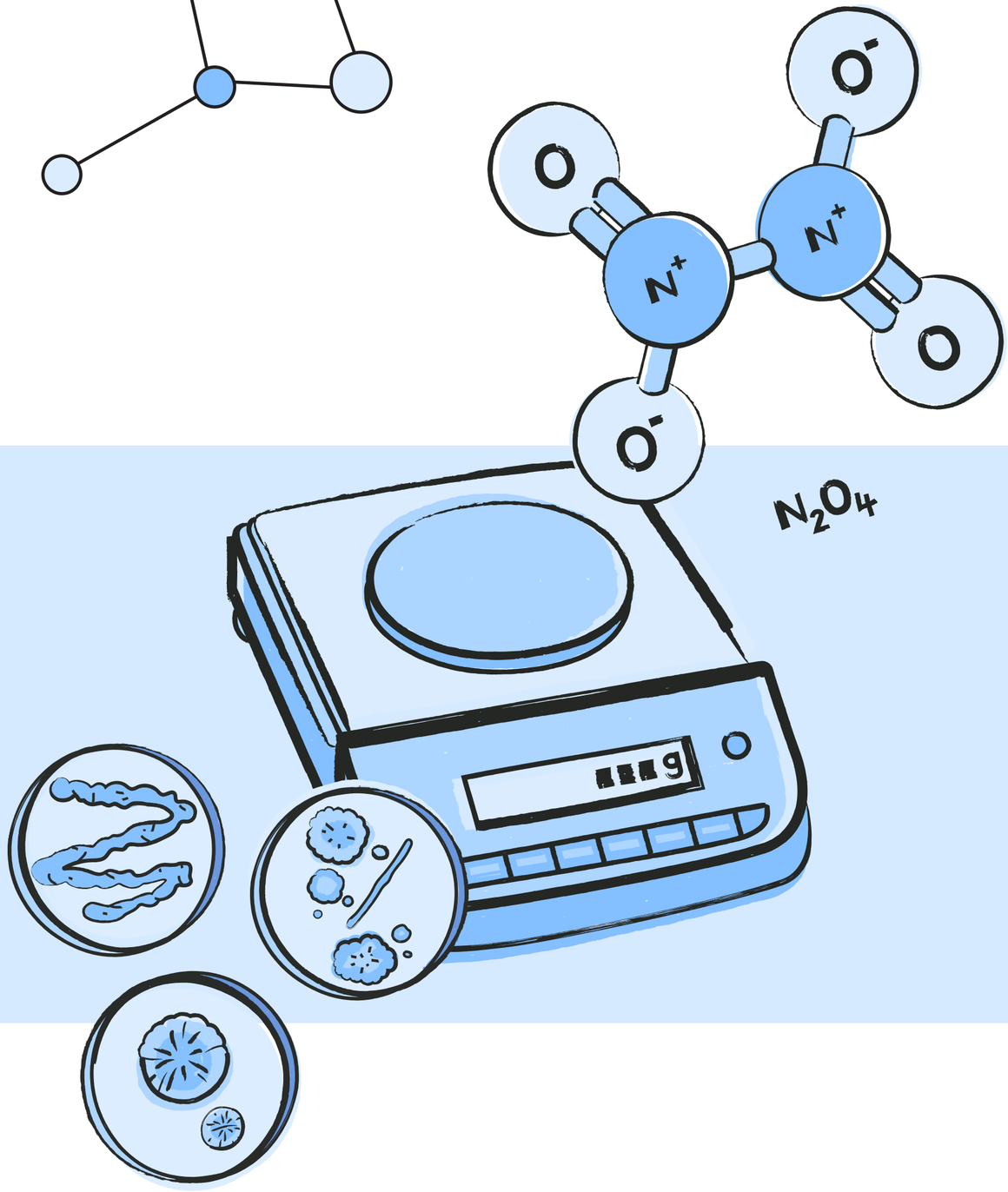




INTRODUCTION & BACKGROUND

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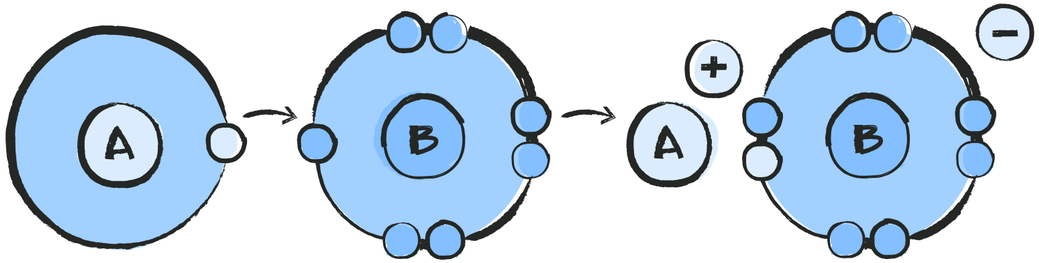


RESEARCH PROBLEM

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RESEARCH OBJECTIVES



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RESEARCH QUESTIONS

QUESTION 1

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QUESTION 2

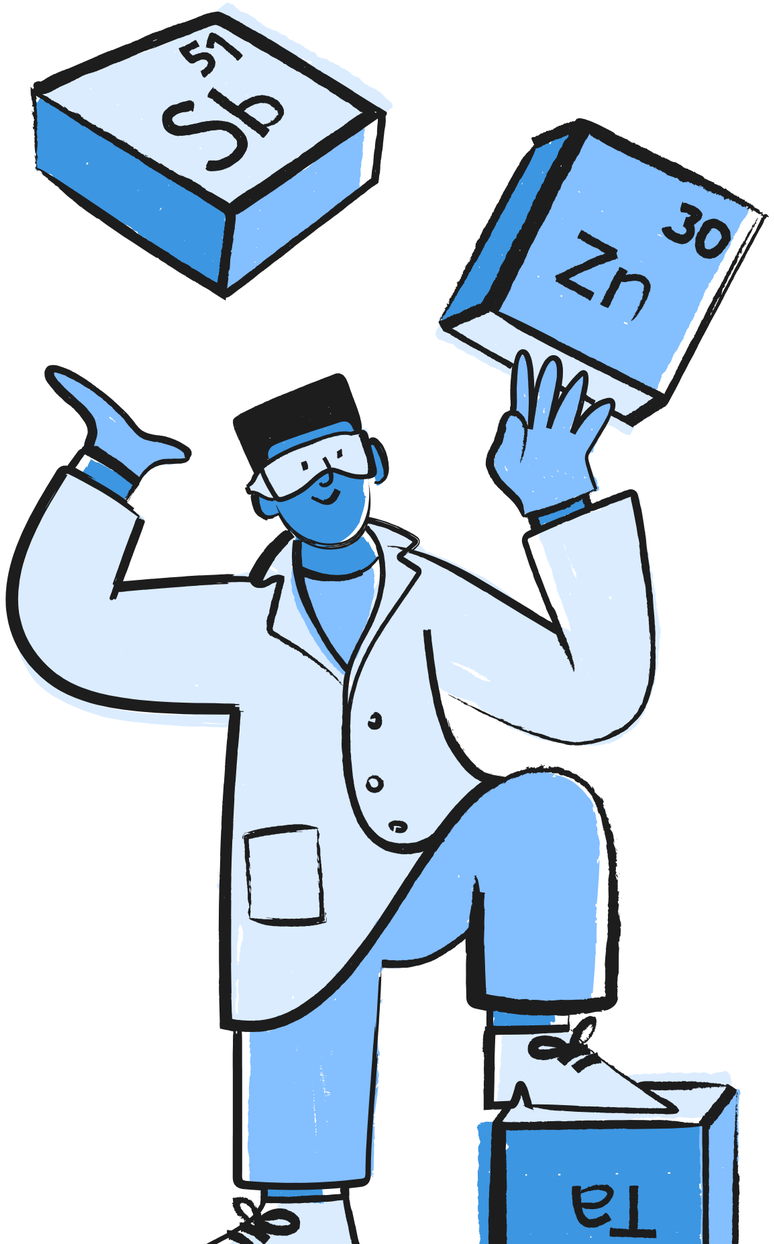
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QUESTION 3

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LITERATURE REVIEW

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RESEARCH METHODOLOGY



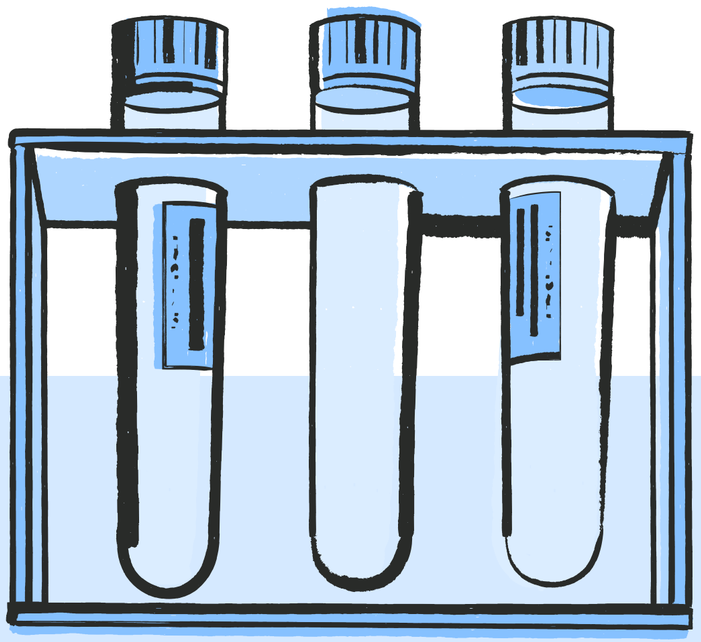
SIGNIFICANCE OF THE STUDY

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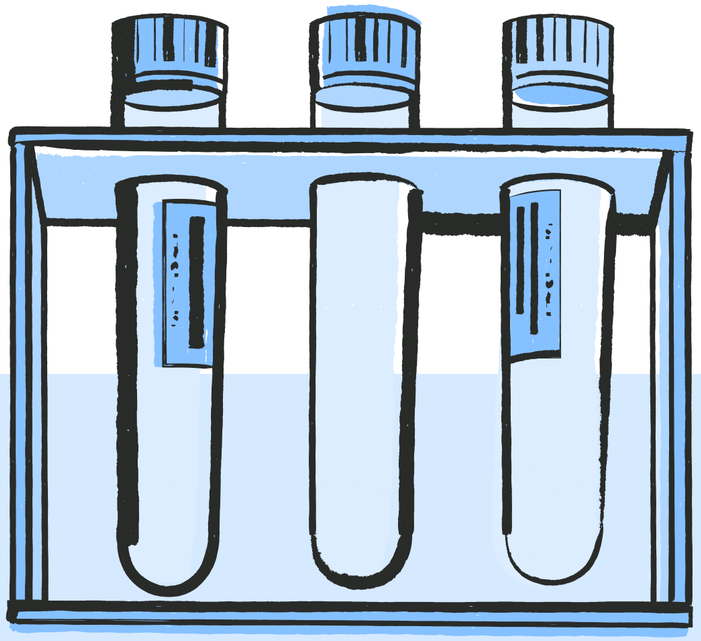
ML & EVALUATION

- Primary success criterion: AUROC ≥ 0.80 on the held-out test set.
- AUPRC should be clearly higher than the prevalence-based baseline.
- Supplementary metrics: Sensitivity, Specificity, F1-score & Brier score
  Assess classification performance at selected thresholds and the reliability of predicted probabilities.
- Interpretability: SHAP
  Identify the clinical features that contribute most strongly to model predictions.



EXPECTED OUTCOMES

- Predictive Feasibility: Assess whether ICU data collected within 6 hours prior to extubation can predict failure within 48 hours after extubation.
- Model Comparison: Compare three models–LightGBM, XGBoost, and Transformer–to identify the one with the best performance.
- Key Clinical Features: Identify the most influential preoperative variables using SHAP analysis.
- Expected Performance: Based on the above evaluation criteria, the top-performing model is expected to demonstrate strong discriminatory ability.



TIMELINE

JANUARY

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FEBRUARY

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MARCH

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REFERENCES

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THANK YOU!

